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# Restoring the Properties of Transparent Al-doped ZnO Thin Film Electrodes Exposed to Ambient Air

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## Abstract

The properties of aluminium doped zinc oxide (AZO) films are known to degrade with exposure to humidity. Different AZO films deposited using reactive direct current (DC) magnetron sputtering and high power impulse magnetron sputtering (HiPIMS) have been aged in ambient conditions and annealed at a low temperature of 160 °C to 180 °C in $N_2$ atmosphere. Their electrical and optical properties have been investigated both ex situ and in situ during the annealing. The electrical and optical properties of the films are improved by the annealing. The results of the in situ measurements are interpreted in terms of a diffusion process, where $Zn(OH)_2$ is decomposed and water is diffusing out of the films. As $Zn(OH)_2$ is known to act as a trap for charge carriers in ZnO, its removal from the film can explain the improvement of the electrical properties by the annealing.

## Introduction

Aluminium doped zinc oxide (AZO) is a transparent conducting oxide. It is often used as a transparent electrode in solar cells.[1] In order to be used as a transparent electrode, AZO films need to have a resistivity in the order of $10^{-4}$ Ωcm, while being transparent in the visible range. AZO films can be produced by various methods, such as magnetron sputtering,[2] spray pyrolysis[3] or chemical vapor deposition.[4] Recently, the possibility of depositing highly conductive AZO films at low temperature has been shown using high power impulse magnetron sputtering.[5] Such low deposition temperatures are interesting for transparent electronics on flexible substrates.[6] However, AZO films have a problem with the degradation of the electrical properties in humid environments.[7]

There have been several studies of this phenomenon,[8–10] that relate the degradation of the electrical properties to the absorption of water and the formation of zinc hydroxide ($Zn(OH)_2$). The hydroxyl groups at the grain boundaries can act as trap states which decrease the charge carrier concentration.[10] The increase in the trap states at the grain boundaries also induces an increase in the potential barrier at the grain boundaries, thus decreasing the mobility of charge carriers.[10,11]

Tohsophon et al.[8] showed that annealing the ZnO films in vacuum at a temperature above 150 °C can partially restore the electrical properties after a damp heat treatment. This suggests that compounds formed after reaction with water may decompose and absorbed water can diffuse out of the film during a low temper-

ature annealing. Hüpkes et al. showed that a high temperature annealing at 650 °C can cause the ZnO films to become stable towards damp heat conditions.[9] They attributed this stability to the reconstruction of the grain boundaries, that serve as diffusion paths for $H_2O$ molecules. Hence, grain boundaries may act as diffusion shortcuts for water in ZnO.

The diffusion of water in oxides has been studied by Doremus.[12] He proposes a model where the diffusion of water is coupled with the chemical reaction of the formation of the hydroxide species to give an effective diffusion coefficient. The diffusion of water is rather difficult to measure directly but its detrimental influence on the electrical and optical properties may enable the development of a method to evidence it indirectly from measuring these physical properties upon water uptake or outflow.

In the present work AZO films prepared by various sputtering processes have been aged under ambient conditions and then annealed under $N_2$ at a temperature between 160 °C and 180 °C. The evolution of the electrical and optical properties of the films has been characterized in situ during the annealing. Additionally, information about the chemical composition and the bonding state of the samples has been obtained before and after the annealing. Results of the in situ analyzes are interpreted in terms of a diffusion process of absorbed water-related species, resulting from the decomposition of $Zn(OH)_2$, out of the AZO films.

## Experimental

Thin films of AZO have been deposited using an alloyed $Zn_{0.97}Al_{0.03}$ target using reactive direct current (DC) magnetron sputtering and reactive high power impulse magnetron sputtering (HiPIMS) onto unheated glass substrates. For the DC magnetron sputtered films three different pressures (0.3 Pa, 1 Pa, 1.5 Pa) were used. The oxygen flow rate was optimized to obtain conductive films as reported by Jullien et al.[2] and a current of 0.1 A was applied. Additionally, AZO films have been deposited by HiPIMS in a different deposition system at a pressure of 1 Pa. Pulses with a length of 100 µs with a frequency of 1000 Hz were supplied by a Melec Spik 2000A. An Advanced Energy Pinnacle Plus DC power supply was used to charge the HiPIMS unit at a constant voltage of 570 V. In this system the films have been placed on a rotating substrate holder, with the magnetron placed excentric to the substrate holder rotation axis. This setup allows to study the film properties as a function of the position of the substrate with respect to the magnetron. The setup is described in more detail by Mickan et al.[5]

The films have been stored in ambient atmosphere for at least 6 months. Then the films have been annealed at a temperature between 160 °C and 180 °C under nitrogen flow using a Linkam FTIR 600 stage and a Linkam THMS 600 stage for in situ optical and electrical characterisation, respectively. The annealing time was between 1.5 hours and 3 hours. Optical characterisation was done with a Cary 5000 UV–Vis–NIR spectrophotometer in the ultraviolet, visible and near-infrared ranges and with a Thermo Scientific Nicolet 6700 Fourier transform infrared (FTIR) spectrometer in the near-infrared range. For the electrical characterisation a 4-point probe setup was used both in situ and ex situ.

Additional characterisation was performed ex situ before and after the annealing. The charge carrier concentration and the mobility were measured using a HMS 5000 Hall effect measurement setup. The composition profile of the films was measured using a CAMECA IMS 7f secondary ion mass spectrometer (SIMS). Photoluminescence (PL) spectra have also been recorded using a Horiba iHR320 spectrometer equipped with a Horiba Syncerity multichannel CCD detector. The excitation source was a 266 nm CryLas FQCW266-50 laser. Two different gratings have been used that are optimized to record the spectra in the UV and visible ranges, respectively.

X-ray photoelectron spectra (XPS) were measured using non-monochromatized $AlK_\alpha$ radiation at 1486.6 eV. The photoelectrons were collected by an Omicron EA 125 hemispherical analyzer. In order to analyze the chemical com-

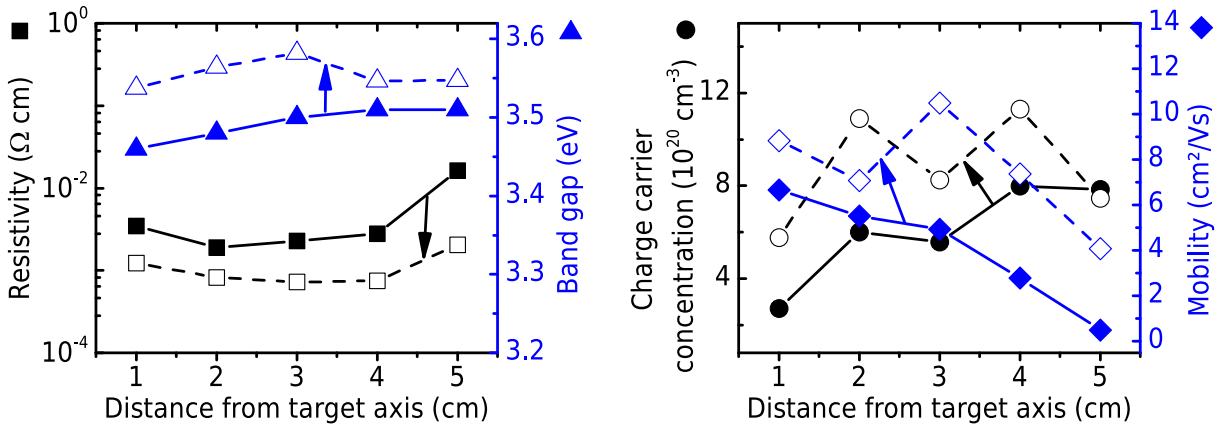



Figure 1: Resistivity, optical band gap, charge carrier concentration and mobility of AZO films before (full symbols) and after (open symbols) an annealing at 180 °C

position of the sample body, the samples were etched by several cycles of $Ar^+$ bombardment ($3 \times 10^{-6}$ mbar) with an energy of 5 keV and a current of 38 mA. The $Ar^+$ bombardment was performed prior to and after annealing at 175 °C for 90 minutes in vacuum.

## Results

The resistivity, charge carrier concentration and mobility have been measured for an AZO sample before and after an annealing at 180 °C for 2 hours. Additionally, the band gap has been calculated from the transmittance spectra using the Tauc method.[13] This AZO sample has been deposited using HiPIMS at a discharge voltage of 570 V. The deposition parameters were chosen to achieve low resistivity values homogeneously across the sample surface. The optical and electrical properties are shown in figure 1 as a function of the lateral position of the sample in the sputtering chamber relative to the center of the target. The resistivity decreases by a factor of 2-8 and the band gap increases. The charge carrier concentration and mobility also increase for most of the sample positions despite the low annealing temperature and the inert atmosphere. Under these annealing conditions changes to the intrinsic defects or the crystalline structure are not expected. This suggests the influence of an extrinsic factor such as absorbed water. In table 1 the resistivity $\rho$, the charge carrier concentration $n$, the mobility $\mu$, the band gap $E_{\mathrm{gap}}$ and the average transmittance in the visible range $T_{\mathrm{avg}}$ are shown for the AZO film deposited at a distance of 3 cm from the target axis. The electrical properties are improving and the average transmittance in the visible range is increasing after the annealing.

**Table 1: Electrical and optical properties of an AZO film deposited using HiPIMS at a distance of 3 cm from the target axis**

| | before annealing | after annealing |
|---|---|---|
| $\rho$ ($10^{-4}\,\Omega$cm) | 22.8 | 7.21 |
| $n$ ($10^{20}\,\mathrm{cm}^{-3}$) | 5.57 | 8.24 |
| $\mu$ ($\mathrm{cm}^2/\mathrm{Vs}$) | 4.93 | 10.5 |
| $E_{\mathrm{gap}}$ (eV) | 3.50 | 3.58 |
| $T_{\mathrm{avg}}$ (%) | 77.0 | 81.5 |

The transmittance of the HiPIMS deposited films has been measured in situ during the annealing at 180 °C in the spectrophotometer. The resulting spectra close to the band gap are shown in figure 2 for the sample deposited at a distance of 3 cm from the target axis. It can be seen that the band gap increases with the time during the annealing. This increase can be related to an increase in the charge carrier density due to the Burstein-Moss effect.[14]

Another indirect way to obtain information about the charge carrier density is the bulk plasmon resonance. The bulk plasmon resonance causes a decrease of the transmittance in the near-infrared range. An increase in the charge

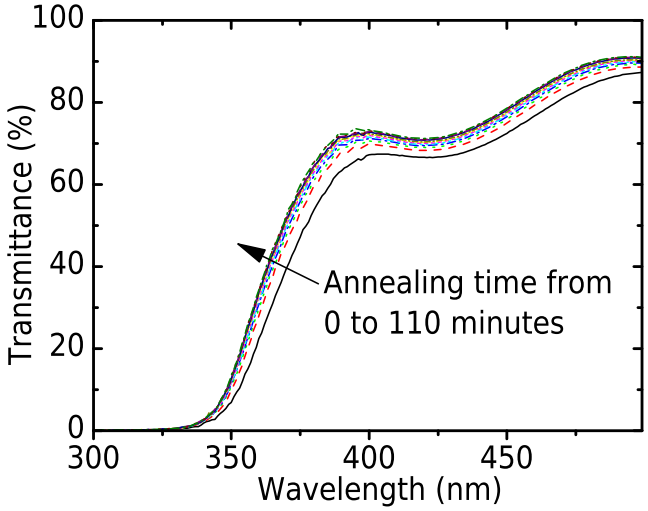



Figure 2: Transmittance of films measured during annealing at 180 °C under $N_2$ as a function of time

carrier concentration leads to a shift of the plasmon resonance and therefore the onset of the decrease in transmittance towards lower wavelengths.[15] This means, that the evaluation of the transmittance at a certain wavelength (e.g. 2500 nm) can be used as an indirect indicator of the charge carrier concentration. The evolution of the transmittance in the near-infrared during the annealing is shown in figure 3a for two different samples. One of the samples is the same HiPIMS deposited sample as in figure 2, which was annealed at 180 °C while measuring its transmittance in the spectrophotometer. The other sample is an AZO film deposited using DC at a pressure of 1.5 Pa. The DC deposited AZO film was annealed at 160 °C while measuring its transmittance using the FTIR spectrometer. The near-infrared transmittance for the DC deposited film is higher due to its higher resistivity and therefore lower charge carrier concentration (see also figure 4). For both samples the transmittance in the near-infrared decreases with the annealing time indicating an increase in the charge carrier concentration. Despite the large difference in the electrical and optical properties of the two samples the evolution of the charge carrier concentration with the annealing is very similar. This can be seen in the relative evolution of the near-infrared transmittance with respect to the initial transmittance

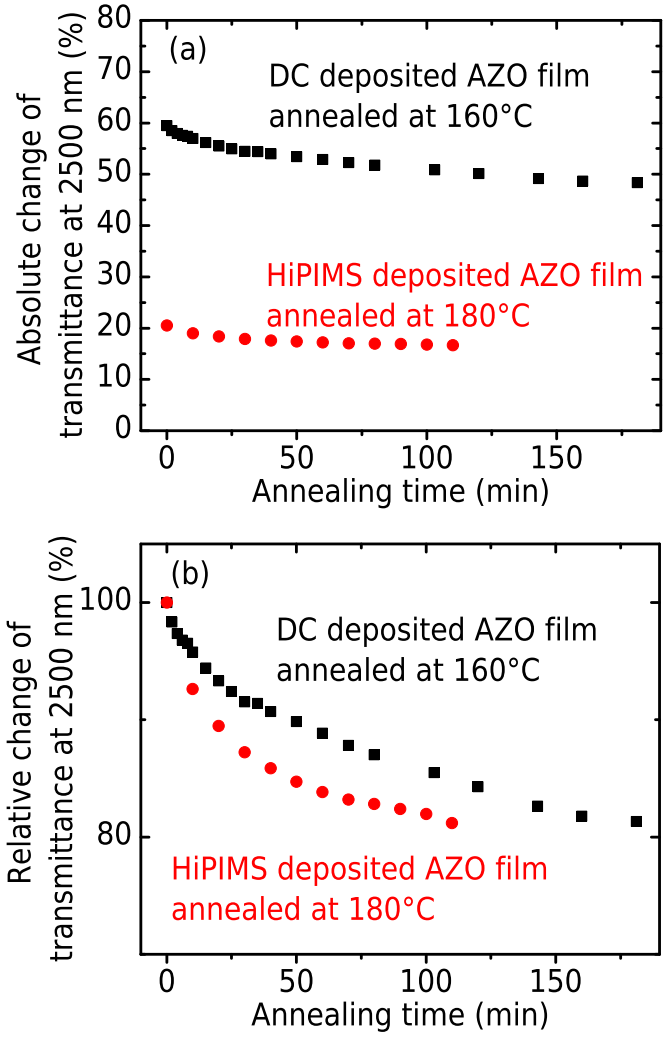
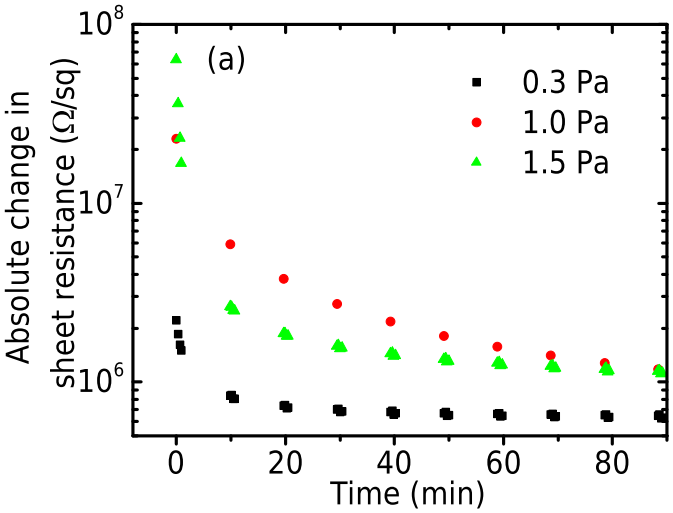



Figure 3: Absolute (a) and relative (b) change of the transmittance at 2500 nm as a function of the annealing time for two different AZO films

that is shown in figure 3b.

Similarly to the optical measurements, in situ resistance measurements have been carried out on AZO films deposited by DC sputtering with different pressures. The evolution of the sheet resistance is shown as a function of the annealing time in figure 4a. Even though the initial sheet resistance of the three samples varies over 2 orders magnitude, the evolution with the annealing time is very similar. The sheet resistance decreases with the annealing time. Figure 4b shows the relative change of the sheet resistance with respect to the initial resistance of each sample. The solid lines show the fit of the function in equation 4 that is explained in the discussion section.

For the HiPIMS deposited sample at a dis-



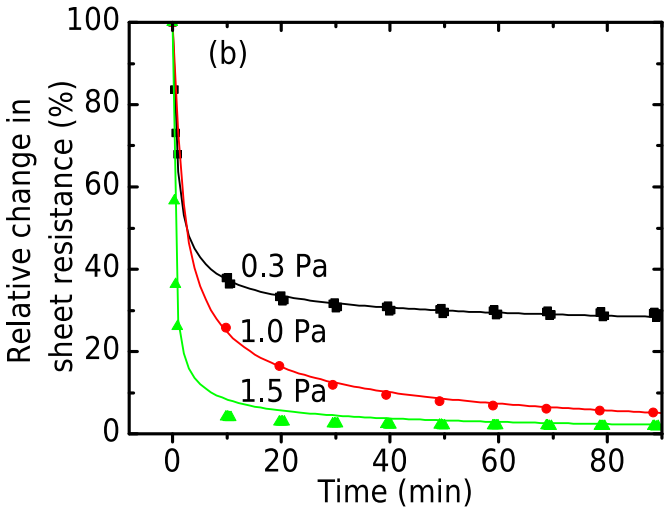




Figure 4: Absolute (a) and relative (b) change of the sheet resistance with the annealing time at a temperature of 170 °C for AZO films deposited using DC sputtering at different pressures. The solid lines in (b) show a fit of the function in equation 4.

tance of 1 cm from the target axis the concentration profile for the elements Zn, Al, O, and H was measured with SIMS before and after annealing at 180 °C for 2 hours. The profiles of most elements did not change considerably with the annealing, except for H. The H profile is shown in figure 5. It can be seen that the H concentration decreases by 65 % after the annealing to a low level throughout the entire film thickness, showing that hydrogen related species have diffused out of the film.

Photoluminescence spectra of an AZO film, that has been deposited on a Si substrate using HiPIMS at a pressure of 0.6 Pa, have been

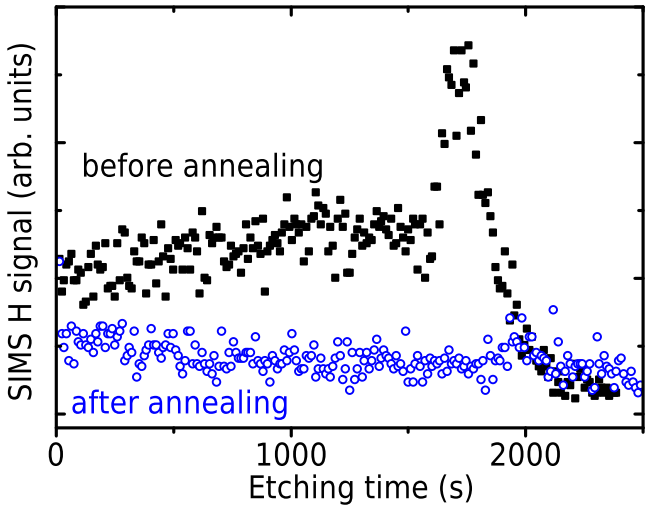



Figure 5: Concentration profile of H in the films before and after annealing

recorded before and after an annealing at 180 °C under $N_2$ for 2 hours. The resulting spectra are shown in figure 6. There is a single peak around 350 nm in the near-band edge region, whose intensity increases with the annealing. The broad peak in the visible range can be attributed to point defects. The intensity of the defect emission decreases slightly after the annealing. It should be noted, that the intensity of the defect emission is much smaller than the intensity of the excitonic emission, even though the emission peaks cannot easily be normalized because two different gratings were used in the different regions of the spectrum.

XPS spectra of AZO films have been measured before and after an annealing in vacuum at 175 °C on two samples that were placed next to each other during the deposition. The films were deposited on Si substrates using DC sputtering at 0.5 Pa. The XPS spectra have been collected on the surface and after several successive Ar bombardment steps. The oxygen 1s core level spectra are shown in figure 7. The O1s peaks were fitted using the software CasaXPS. The background was treated by the Shirley method.[16] The oxygen peak can be fitted by 4 contributions, that are a mixture of Gaussian and Lorentzian functions. The contributions are labeled A,B,C and D. Contribution A is centered around 529.9 eV ± 0.05 eV and can be attributed to the Zn-O bonds in the

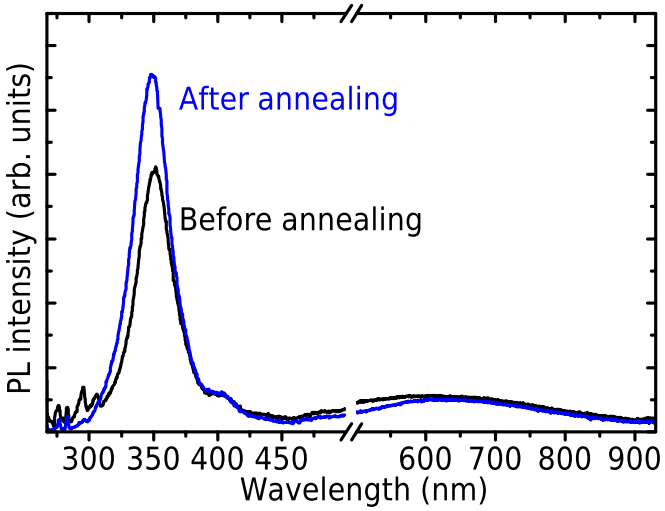



Figure 6: PL spectra of a HiPIMS deposited AZO film before and after annealing at 180 °C under $N_2$

ZnO lattice.[17] A second contribution B (centered around 531.1 nm ± 0.05 eV) can be attributed to Zn-O bonds in oxygen deficient regions in the ZnO lattice.[17] Contribution C (centered around 532.4 eV ± 0.06 eV) can be attributed to the contamination with for example -$CO_3$, water and hydroxyl groups.[17] Contribution D (centered at 533.7 nm ± 0.23 eV) was found by Kunat et al.[18] and confirmed by Kotsis and Staemmler,[19] to be characteristic for hydroxide species on the polar (0001) surface of ZnO. This contribution disappears after the Ar bombardment of the sample.

From the areas of the peaks corresponding to the contributions A to D, the content of oxygen belonging to hydroxide species in the films can be estimated by the ratio (C+D)/(A+B+C+D). This estimation is probably to high, because contribution C is not just attributed to hydroxide, but also to other contaminants. The hydroxide content is shown in figure 8a. It shows that the hydroxide content within the film decreases from around 20 % before the annealing to around 17 % after the annealing. In figure 8b the ratio of A/(A+B) is shown as a function of the bombardment time. This ratio shows the fraction of oxygen within a stoichiometric ZnO environment of the total oxygen attributed to ZnO. After the annealing more of the oxygen belonging to ZnO can be found in a stoichiometric environment.

# Discussion

The results of the in situ characterisation during the annealing show that the electrical properties are improved by the annealing. The transmittance spectra show, that there is a gradual shift of the bulk plasmon wavelength towards higher energy, that corresponds to an increase in the charge carrier concentration.[15] The results of the Hall effect measurements show that also the mobility is improved by the annealing. This improvement of the electrical properties could be explained by the removal of $Zn(OH)_2$. $Zn(OH)_2$ is believed to play a role in the degradation of the electrical properties of ZnO films due to humidity,[7] as it is a wide band gap semiconductor with a band gap that was calculated to be 5.645 eV.[20] $Zn(OH)_2$ is known to decompose at a temperature of 150 °C.[21] Kim et al. have shown that the increase in resistance during a damp heat treatment of AZO can be explained by the chemisorbed $OH^-$ at grain boundaries[10] that forms trap states for charge carriers and increases the potential barrier at grain boundaries. The resistance of the film can be modeled as the sum of the resistance within the grains and the resistance due to the grain boundaries. The $OH^-$ groups at the grain boundaries can trap charge carriers and therefore increase the grain boundary resistance.

It can be assumed, that the $Zn(OH)_2$ is decomposed and diffusing out of the grain boundaries in the form of molecular water during the annealing.[12] The change of the resistance with the annealing time in figure 4 can therefore be interpreted in terms of a diffusion process of the molecular water. In order to model this diffusion process, it is assumed that the thin film is a plane source for the diffusing substance. The diffusion from a plane source where the total amount of a substance $M_0$ is located at the position $x = 0$ at a time $t = 0$ can be described by

$$C = \frac{M_0}{\sqrt{\pi D t}} \exp(-x^2/4Dt) \ , \qquad (1)$$

with $D$ being the diffusion coefficient and $C$ the

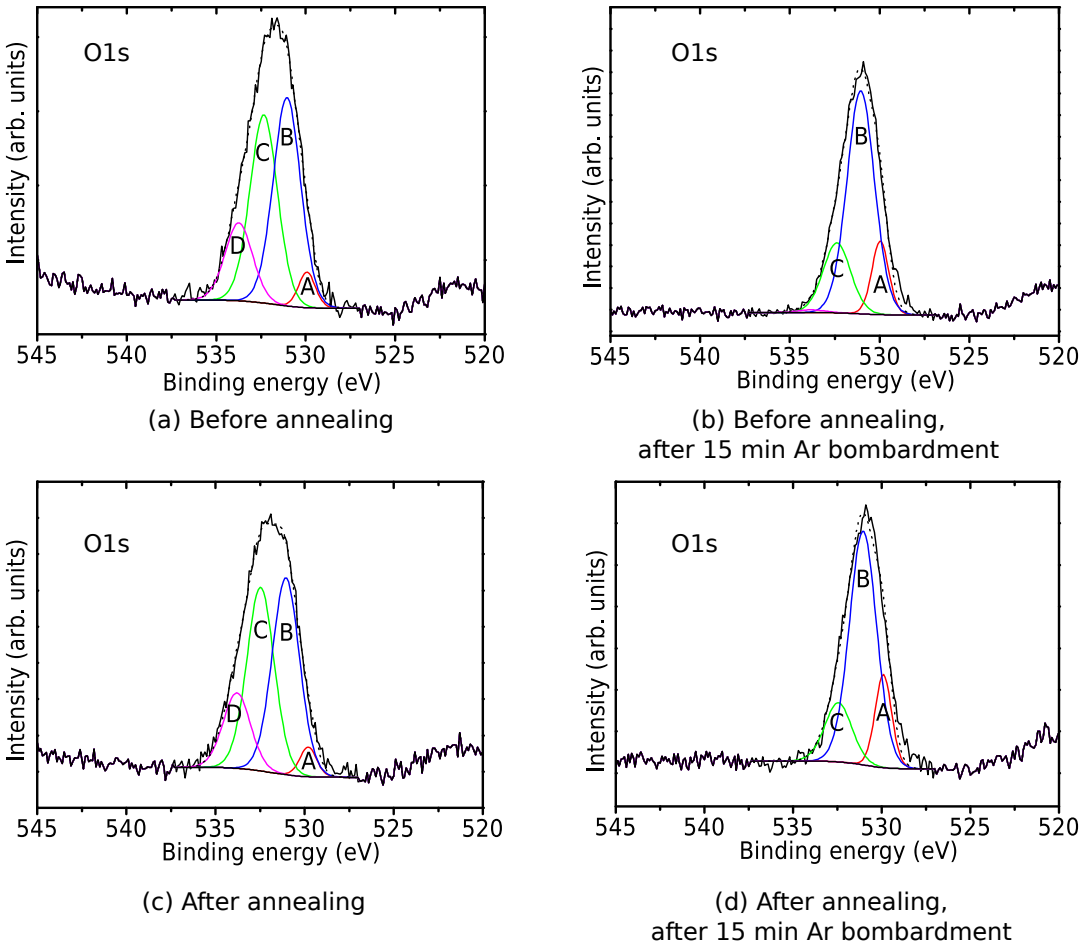



Figure 7: XPS spectra before (a,b) and after (c,d) annealing at 175 °C

linear concentration.[22] The amount $M(t)$ of the substance that stays in the film of thickness $h$ after a time $t$ can be calculated by integrating over equation 1.

$$M(t) = \int_0^h C \, \mathrm{d}x = M_0 \, \mathrm{erf}\frac{h}{2\sqrt{Dt}} \qquad (2)$$

It is possible to normalize the measured resistance $R(t)$ of the films, in order to obtain a quantity that is proportional to the amount $M(t)$ with $M_0$ equal to 1. This normalization should be between 100 % and a lower bound $C_0$ and can be described by

$$R_{\mathrm{normalized}} = \frac{R(t) - R_{\mathrm{min}}}{R_{\mathrm{max}} - R_{\mathrm{min}}}(1 - C_0) + C_0 \ , \qquad (3)$$

with $R_{\mathrm{max}}$ the initial (maximum) value of the resistance and $R_{\mathrm{min}}$ the final (minimum) value of the resistance. The lower bound $C_0$ corresponds to the concentration of the substance $M$ at the end of the annealing. Considering that the substance $M$ is not necessarily completely removed from the films during the annealing, $C_0$ can be assumed to be larger than 0. One could also assume that $C_0$ corresponds to the ratio of the lowest to the highest resistance value measured ($R_{\mathrm{min}}/R_{\mathrm{max}}$), however the resistance of the films depends on many other factors such as grain boundaries or impurity scattering, so $C_0$ should be independent of $R_{\mathrm{min}}/R_{\mathrm{max}}$. By replacing $R_{\mathrm{normalized}}$ with $M(t)$ from equation 2, equation 3 can be rewritten to be able to fit it to the measured relative change of the resistance $R(t)/R_{\mathrm{max}}$:

$$\frac{R(t)}{R_{\mathrm{max}}} = \frac{M(t) - C_0}{1 - C_0}(1 - \frac{R_{\mathrm{min}}}{R_{\mathrm{max}}}) + \frac{R_{\mathrm{min}}}{R_{\mathrm{max}}} \qquad (4)$$

Using equations 2 and 4, there are two parameters for the fit, the diffusion coefficient $D$ and the normalization parameter $C_0$. The resulting curves are shown in figure 4b. The results for the diffusion coefficient $D$ and the normaliza-

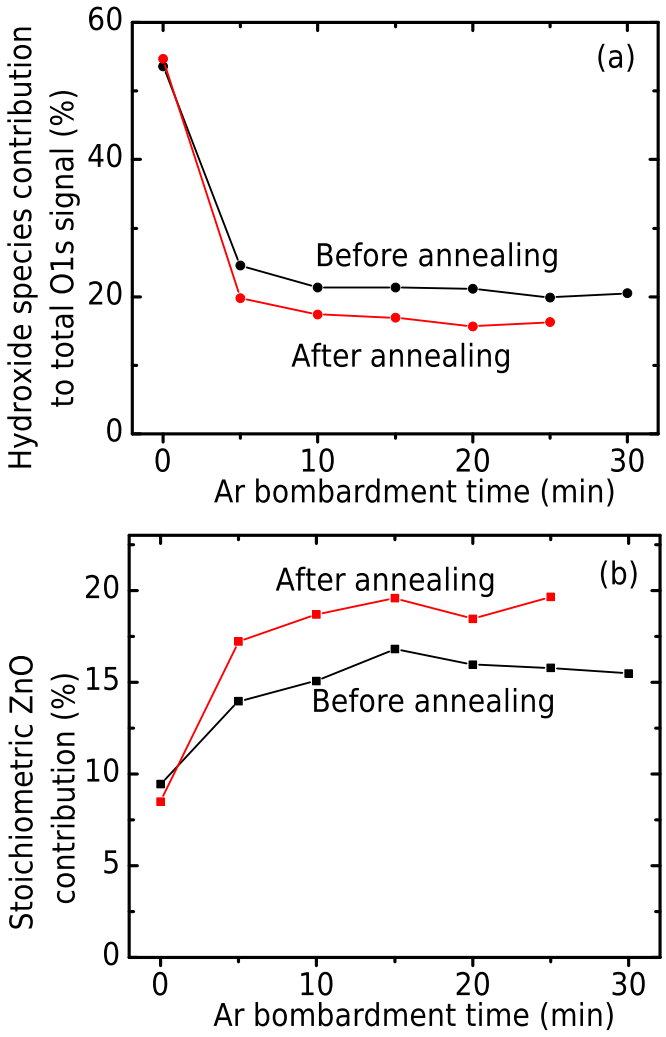



Figure 8: Ratio of the fitting contributions of the O1s XPS signal. (a) Contribution of oxygen belonging to hydroxide species. (b) Contribution of oxygen within the stoichiometric ZnO lattice

tion parameter $C_0$ are summarized in table 2.

**Table 2: Fit parameters of equation 4 for the measured resistance**

| Pressure (Pa) | $D$ ($10^{-12}$cm$^2$/s) | $C_0$ (%) |
|---|---|---|
| 0.3 | 1.7 | 6.3 |
| 1.0 | 0.6 | 8.5 |
| 1.5 | 6.3 | 2.3 |

The fitted diffusion coefficient has the highest value for the sample deposited with the highest pressure. A high deposition pressure results in more porous films according to the structure zone model by Thornton.[23] A more porous film allows for faster diffusion. This suggests that the hydroxide in the grain boundaries might decompose and water might be diffusing out of the material using the grain boundaries as diffusion paths.

The SIMS results confirm that hydrogen is removed from the films during the annealing. This is in agreement with the model of the decomposition of $Zn(OH)_2$. The increase in the PL intensity of the excitonic peak is also in agreement with the hypothesis of the decomposition of $Zn(OH)_2$. It has been shown that the presence of $ZnOH_2$ decreases the intensity of the excitonic emission.[24]

The fitting results of the O1s core level XPS peak also show that the concentration of hydroxide species is decreasing after the annealing. At the same time the concentration of oxygen in a stoichiometric ZnO lattice is increased. This could indicate, that when the hydroxide is removed from the film, the environment around those oxygen atoms changes to resemble more the environment of stoichiometric ZnO according to

$$Zn(OH)_2 \rightarrow ZnO + H_2O \ . \qquad (5)$$

The water can then diffuse out of the film along the grain boundaries, as discussed above.

# Conclusion

Various AZO films have been annealed under $N_2$ atmosphere at 160 °C to 180 °C. The evolution of their electrical and optical properties has been followed in situ as a function of the annealing time. The electrical and optical properties improve after the annealing in a similar way, independent of the initial properties. The improvement of the electrical properties can be related to a diffusion process, where the fastest diffusion is found for films deposited at high pressure. SIMS measurements have shown that hydrogen is removed from the films. XPS O1s core level spectra show that there is a decrease in the content of hydroxyl groups after the annealing. This shows, that $Zn(OH)_2$ is decomposed and water is removed from the films during the annealing.

# References


(1) Beyer, W.; Hüpkes, J.; Stiebig, H. Transparent conducting oxide films for thin film silicon photovoltaics. *Thin Solid Films* **2007**, *516*, 147–154.

(2) Jullien, M.; Horwat, D.; Manzeh, F.; Escobar Galindo, R.; Bauer, P.; Pierson, J.; Endrino, J. Influence of the nanoscale structural features on the properties and electronic structure of Al-doped ZnO thin films: An X-ray absorption study. *Solar Energy Materials and Solar Cells* **2011**, *95*, 2341–2346.

(3) Inamdar, D.; Agashe, C.; Kadam, P.; Mahamuni, S. Doping optimization and surface modification of aluminum doped zinc oxide films as transparent conductive coating. *Thin Solid Films* **2012**, *520*, 3871–3877.

(4) Volintiru, I.; Creatore, M.; Kniknie, B. J.; Spee, C. I. M. A.; van de Sanden, M. C. M. Evolution of the electrical and structural properties during the growth of Al doped ZnO films by remote plasma-enhanced metalorganic chemical vapor deposition. *Journal of Applied Physics* **2007**, *102*, 043709.

(5) Mickan, M.; Helmersson, U.; Rinnert, H.; Ghanbaja, J.; Muller, D.; Horwat, D. Room Temperature Deposition of Homogeneous, Highly Transparent and Conductive Al-Doped ZnO Films by Reactive High Power Impulse Magnetron Sputtering. *Solar Energy Materials and Solar Cells* **2016**, *157*, 742–749.

(6) Fortunato, E.; Barquinha, P.; Martins, R. Oxide Semiconductor Thin-Film Transistors: A Review of Recent Advances. *Advanced Materials* **2012**, *24*, 2945–2986.

(7) Pern, F.; Noufi, R.; Li, X.; DeHart, C.; To, B. Damp-heat induced degradation of transparent conducting oxides for thin-film solar cells. 33rd IEEE Photovoltaic Specialists Conference, 2008. PVSC '08. 2008; pp 1–6.

(8) Tohsophon, T.; Hüpkes, J.; Calnan, S.; Reetz, W.; Rech, B.; Beyer, W.; Sirikulrat, N. Damp heat stability and annealing behavior of aluminum doped zinc oxide films prepared by magnetron sputtering. *Thin Solid Films* **2006**, *511–512*, 673–677.

(9) Hüpkes, J.; Owen, J.; Wimmer, M.; Ruske, F.; Greiner, D.; Klenk, R.; Zastrow, U.; Hotovy, J. Damp heat stable doped zinc oxide films. *Thin Solid Films* **2014**, *555*, 48–52.

(10) Kim, J. I.; Lee, W.; Hwang, T.; Kim, J.; Lee, S.-Y.; Kang, S.; Choi, H.; Hong, S.; Park, H. H.; Moon, T. et al. Quantitative analyses of damp-heat-induced degradation in transparent conducting oxides. *Solar Energy Materials and Solar Cells* **2014**, *122*, 282–286.

(11) Steinhauser, J.; Meyer, S.; Schwab, M.; Faÿ, S.; Ballif, C.; Kroll, U.; Borrello, D. Humid Environment Stability of Low Pressure Chemical Vapor Deposited Boron Doped Zinc Oxide Used as Transparent Electrodes in Thin Film Silicon Solar Cells. *Thin Solid Films* **2011**, *520*, 558–562.

(12) Doremus, R. H. Diffusion of water in crystalline and glassy oxides: Diffusion–reaction model. *Journal of Materials Research* **1999**, *14*, 3754–3758.

(13) Tauc, J.; Grigorovici, R.; Vancu, A. Optical Properties and Electronic Structure of Amorphous Germanium. *physica status solidi (b)* **1966**, *15*, 627–637.

(14) Sernelius, B. E.; Berggren, K.-F.; Jin, Z.-C.; Hamberg, I.; Granqvist, C. G. Band-gap tailoring of ZnO by means of heavy Al doping. *Physical Review B* **1988**, *37*, 10244–10248.

(15) Chopra, K. L.; Major, S.; Pandya, D. K. Transparent conductors—A status review. *Thin Solid Films* **1983**, *102*, 1–46.

(16) Shirley, D. A. High-Resolution X-Ray Photoemission Spectrum of the Valence Bands of Gold. *Physical Review B* **1972**, *5*, 4709–4714.

(17) Chen, M.; Wang, X.; Yu, Y. H.; Pei, Z. L.; Bai, X. D.; Sun, C.; Huang, R. F.; Wen, L. S. X-Ray Photoelectron Spectroscopy and Auger Electron Spectroscopy Studies of Al-Doped ZnO Films. *Applied Surface Science* **2000**, *158*, 134–140.

(18) Kunat, M.; Girol, S. G.; Burghaus, U.; Wöll, C. The Interaction of Water with the Oxygen-Terminated, Polar Surface of ZnO. *The Journal of Physical Chemistry B* **2003**, *107*, 14350–14356.

(19) Kotsis, K.; Staemmler, V. Ab Initio Calculations of the O1s XPS Spectra of ZnO and Zn Oxo Compounds. *Physical Chemistry Chemical Physics* **2006**, *8*, 1490–1498.

(20) Wang, M.; Jiang, L.; Kim, E. J.; Hahn, S. H. Electronic structure and optical properties of Zn(OH)2: LDA+U calculations and intense yellow luminescence. *RSC Advances* **2015**, *5*, 87496–87503.

(21) Shaporev, A. S.; Ivanov, V. K.; Baranchikov, A. E.; Polezhaeva, O. S.; Tret'yakov, Y. D. ZnO formation under hydrothermal conditions from zinc hydroxide compounds with various chemical histories. *Russian Journal of Inorganic Chemistry* **2007**, *52*, 1811–1816.

(22) Crank, J. *The mathematics of diffusion*, 2nd ed.; Clarendon Press: Oxford, [Eng], 1975.

(23) Thornton, J. A. Influence of apparatus geometry and deposition conditions on the structure and topography of thick sputtered coatings. *Journal of Vacuum Science & Technology* **1974**, *11*, 666–670.

(24) Zhou, H.; Alves, H.; Hofmann, D.; Meyer, B.; Kaczmarczyk, G.; Hoffmann, A.; Thomsen, C. Effect of the (OH) Surface Capping on ZnO Quantum Dots. *physica status solidi (b)* **2002**, *229*, 825–828.